Exponentials & Logarithms

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $\ln y = 2\ln x + \ln 3$, express y in terms of x. Hence sketch the graph.

$\ln y = \ln x^2 + \ln 3 = \ln(3x^2) \Rightarrow y = 3x^2$. Parabola through origin, scaled by 3.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Advanced Trigonometry

Solve the separable DE $\frac{dy}{dx} = \frac{x+1}{y}$ given $y = 2$ when $x = 0$.

Separate: $y\,dy = (x+1)\,dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + x + c$. At $(0, 2)$: $2 = c$. So $y^2 = x^2 + 2x + 4$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Differentiation

Given that $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$, find $\frac{z_1}{z_2}$ in the form $a + bi$.

Multiply by conjugate: $\frac{(3+4i)(1+2i)}{(1-2i)(1+2i)} = \frac{3+6i+4i-8}{1+4} = \frac{-5+10i}{5} = -1 + 2i$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Integration

A curve is defined parametrically by $x = t^2 + 1$, $y = 2t^3 - 3t$. Find $\frac{dy}{dx}$ in terms of t and the value at $t = 1$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6t^2 - 3$. $\frac{dy}{dx} = \frac{6t^2-3}{2t}$. At $t = 1$: $\frac{3}{2}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Sequences & Series

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

Sketch: V-shape with vertex at $(1.5, 0)$. Solve: $2x - 3 = 5 \Rightarrow x = 4$ or $2x - 3 = -5 \Rightarrow x = -1$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Vectors

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \cdots = 1 + 2x + 2x^2 + \cdots$. Valid for all $x \in \mathbb{R}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$ where $R > 0$ and $0 < \alpha < 90°$.

$R = \sqrt{9 + 16} = 5$. $\tan\alpha = 4/3 \Rightarrow \alpha = 53.13°$. So $5\sin(\theta + 53.13°)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the equation $e^{2x-1} = 5$, giving your answer in exact form.

a. $x = \frac{1+\ln 5}{2}$ ✓

b. $x = \frac{\ln 5 - 1}{2}$

c. $x = \ln\left(\frac{5}{2}\right) + 1$

d. $x = \frac{\ln 6}{2}$

Take natural logarithms of both sides: $2x - 1 = \ln 5 \Rightarrow 2x = 1 + \ln 5 \Rightarrow x = \frac{1+\ln 5}{2}$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the expression $e^{3\ln x} + \ln(e^{2x})$ for $x > 0$.

a. $x^3 + 2x$ ✓

b. $3x + 2x = 5x$

c. $x^3 + e^{2x}$

d. $3x^2 + 2$

$e^{3\ln x} = e^{\ln(x^3)} = x^3$. $\ln(e^{2x}) = 2x$. Sum is $x^3 + 2x$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the equation $2^{x+1} = 3^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 2}{\ln 3 - \ln 2}$ ✓

b. $x = \frac{\ln 3}{\ln 3 - \ln 2}$

c. $x = \frac{\ln 2}{\ln 3 + \ln 2}$

d. $x = \frac{1}{\ln(3/2)}$

$\ln(2^{x+1}) = \ln(3^x) \Rightarrow (x+1)\ln 2 = x\ln 3 \Rightarrow x\ln 2 + \ln 2 = x\ln 3 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the equation $e^x - 6e^{-x} = 1$ for real x.

a. $x = \ln 3$ ✓

b. $x = \ln 3$ or $x = \ln(-2)$

c. $x = \ln 2$

d. $x = 3$

Multiply by e^x: $(e^x)^2 - e^x - 6 = 0 \Rightarrow (e^x - 3)(e^x + 2) = 0$. Since $e^x > 0$, $e^x = 3 \Rightarrow x = \ln 3$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $5^{2x-1} = 7$, giving your answer correct to 3 significant figures.

a. 1.10 ✓

b. 0.852

c. 1.21

d. 0.945

$(2x-1)\ln 5 = \ln 7 \Rightarrow 2x - 1 = \frac{\ln 7}{\ln 5} = \frac{1.9459}{1.6094} \approx 1.209 \Rightarrow 2x \approx 2.209 \Rightarrow x \approx 1.10$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The variables x and y satisfy $y = ab^x$. When $\ln y$ is plotted against x, a straight line is obtained. What are the gradient and vertical intercept?

a. **Gradient** $= \ln b$**, Intercept** $= \ln a$ ✓

b. Gradient $= b$, Intercept $= a$

c. Gradient $= \ln a$, Intercept $= \ln b$

d. Gradient $= a$, Intercept $= \ln b$

Take logarithms: $\ln y = \ln(a \cdot b^x) = \ln a + x\ln b = (\ln b)x + \ln a$. This matches $Y = mX + c$ with gradient $\ln b$ and intercept $\ln a$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The variables x and y satisfy $y = kx^n$. The line of $\ln y$ against $\ln x$ has gradient 2.5 and intercept $\ln y = 1.386$. Find k and n.

a. $k = 4, n = 2.5$ ✓

b. $k = 2.5, n = 4$

c. $k = 1.386, n = 2.5$

d. $k = e^{2.5}, n = 1.386$

$\ln y = n\ln x + \ln k$. Gradient $n = 2.5$. Intercept $\ln k = 1.386 \Rightarrow k = e^{1.386} = 4$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Points $(2, 3.2)$ and $(5, 5.6)$ lie on the straight line of $\ln y$ against x. Find the value of b if $y = ab^x$.

a. $b = e^{0.8} \approx 2.226$ ✓

b. $b = 0.8$

c. $b = 2.4$

d. $b = e^{1.6} \approx 4.95$

Gradient $m = \ln b = \frac{5.6 - 3.2}{5 - 2} = \frac{2.4}{3} = 0.8$. Thus $b = e^{0.8} \approx 2.226$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The mass of a radioactive substance decays according to $M(t) = 80e^{-0.05t}$, where t is in years. Find the half-life of the substance in years.

a. ≈ 13.9 years ✓

b. 20.0 years

c. 10.0 years

d. 15.5 years

$M(t) = 40 \Rightarrow 80e^{-0.05t} = 40 \Rightarrow e^{-0.05t} = 0.5 \Rightarrow -0.05t = -\ln 2 \Rightarrow t = \frac{\ln 2}{0.05} = 20\ln 2 \approx 13.86$ years.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A bacterial culture grows according to $N(t) = 500e^{kt}$. If $N(3) = 2000$, find the exact value of k.

a. $k = \frac{\ln 4}{3}$ ✓

b. $k = \frac{\ln 2}{3}$

c. $k = \frac{4}{3}$

d. $k = \ln\left(\frac{4}{3}\right)$

$500e^{3k} = 2000 \Rightarrow e^{3k} = 4 \Rightarrow 3k = \ln 4 \Rightarrow k = \frac{\ln 4}{3}$.

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A hot beverage cools according to Newton's law of cooling: $T(t) = 20 + 70e^{-0.08t}$, where T is temperature in $^\circ$C and t is time in minutes. How long does it take to cool to $45\,^\circ$C?

a. ≈ 12.9 min ✓

b. 15.2 min

c. 10.5 min

d. 18.0 min

$45 = 20 + 70e^{-0.08t} \Rightarrow 25 = 70e^{-0.08t} \Rightarrow e^{-0.08t} = 25/70 = 5/14 \Rightarrow -0.08t = \ln(5/14) \approx -1.0296 \Rightarrow t \approx 1.0296/0.08 \approx 12.87$ min.

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Given $\log_b 3 = A$, $\log_b 11 = B$ and $\log_b 5 = C$, express $\log_b 25$ in terms of A, B and C.

a. A+2C

b. 2A

c. 3C

d. **2C** ✓

$25 = 3^0 \cdot 11^0 \cdot 5^2 \Rightarrow \log_b 25 = 2C$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $\log_b 3 = A$ and $\log_b 2 = B$, express $\log_b 3$ in terms of A and B.

a. A+B

b. **A ✓**

c. 2A

d. B

$3 = 3^1 \cdot 2^0 \Rightarrow \log_b 3 = A$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $\tan\theta = -\frac{6}{5}$ with θ in Quadrant II, state the sign of $\sec\theta$.

a. positive

b. **negative ✓**

c. undefined

d. zero

Quadrant II: $\cos\theta$ is negative, so $\sec\theta$ is negative.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An arithmetic sequence has first term $a = 1$ and common difference $d = -4$. Find the sum of the first 3 terms.

a. -10

b. -7

c. -13

d. **-9 ✓**

$S_n = \frac{n}{2}(2a + (n-1)d) \Rightarrow S_3 = \frac{3}{2}(2 - 8) = -9$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\tan\theta = 3$ with θ in Quadrant I, state the sign of $\sec\theta$.

a. undefined

b. zero

c. **positive ✓**

d. negative

Quadrant I: $\cos\theta$ is positive, so $\sec\theta$ is positive.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric series has first term $a = 2$ and common ratio $r = -\frac{1}{4}$. Find the sum to infinity.

a. $\frac{8}{5}$ ✓

b. $\frac{18}{5}$

c. $\frac{5}{2}$

d. $-\frac{8}{5}$

$S_\infty = \frac{a}{1-r} = \frac{2}{1-(-\frac{1}{4})} = \frac{8}{5}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Sequences & Series

Find the remainder when $x^2 - 5x - 4$ is divided by $(x - 3)$.

a. -2

b. 10

c. **-10** ✓

d. -9

Dividing: quotient $= x - 2$, remainder $= -10$ (check: $P(3) = -10$).

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Vectors

The locus $|z - (3 - 2i)| = 4$ is a circle. What is its center?

a. (3, -1)

b. **(3, -2)** ✓

c. (4, -2)

d. (-3, 2)

$|z - (3 - 2i)| = 4$ is a circle centered at $(3, -2)$ with radius 4.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Given $y = 5e^{2x}$, find the coefficient of e^{2x} in $\frac{dy}{dx}$.

a. 11

b. **10** ✓

c. 2

d. 7

$\frac{dy}{dx} = 5 \times 2\, e^{2x} = 10e^{2x}$. Coefficient: 10.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Complex Numbers

Solve $\cos(\theta) = -\frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the smallest solution (in degrees).

a. 510

b. 210

c. 151

d. **150** ✓

Solutions: $150^\circ, 210^\circ$. Smallest: 150°.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the point of intersection (with positive imaginary part) of $|z| = 5$ and $\text{Re}(z) = 3$.

a. (3, -4)

b. (4, 3)

c. **(3, 4)** ✓

d. (5, 3)

$y = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$; intersection point $(3, 4)$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Numerical Methods

Solve $\sin(\theta) = \frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the largest solution (in degrees).

a. 119

b. 0

c. 60

d. **120** ✓

Solutions: $60^\circ, 120^\circ$. Largest: 120°.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Differential Equations

A geometric series with common ratio $r = \frac{1}{3}$ has sum to infinity $S_\infty = 12$. Find the first term a.

a. 11

b. 7

c. **8** ✓

d. -8

$a = S_\infty(1 - r) = 12 \times \left(1 - \left(\frac{1}{3}\right)\right) = 8$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Poisson Distribution

Find the magnitude of the vector $\mathbf{v} = \langle 4, -4, 7 \rangle$.

a. 15

b. **9** ✓

c. 8

d. 10

$|\mathbf{v}| = \sqrt{(4)^2 + (-4)^2 + (7)^2} = \sqrt{81} = 9.$

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